Appendix I. Debt Sustainability Analysis[8]

46. **Zimbabwe's debt sustainability analysis (DSA) exercise is subject to major caveats.** While the authorities have begun to shore up their debt management capacity, external debt and debt service data are not reconciled with the creditors' data and remain incomplete. Data deficiencies are compounded by the difficulty in making medium- and long-term projections because of the country's poor policy record. Against this background, the DSA projections and simulations can only be considered as highly indicative.

47. **Zimbabwe is in debt distress.** Even under relatively optimistic assumptions on policies and the external environment, most debt ratios are projected to remain at unsustainable levels over the medium and long term. Under the historic sensitivity analysis, debt ratios are projected to increase over the long term from the current unsustainable levels.

Public and publicly guaranteed external debt sustainability

48. **The starting external debt level is high.** At end-2008, the staff estimates that Zimbabwe's nominal medium- and long-term public and publicly guaranteed (PPG) external debt stood at US$5.1 billion (166 percent of GDP).[9] Scheduled service of PPG external debt in 2009 is equivalent to 14 percent of projected exports and 28 percent of projected government revenue (Table 7).

49. **The baseline scenario assumes improvements in policies and a reversal of Zimbabwe's decade-long economic contraction.** In particular, the scenario is built on implementation of sound macroeconomic policies and essential structural reforms as discussed in Section IV of the staff report. Annual real GDP growth is projected at 6 percent during the period of 2010–15 and 5.5 percent thereafter. The noninterest external current account deficit would initially fall—from 25 percent of GDP in 2010 to 14 percent of GDP in 2015—but would then decline more gradually. It is assumed that private capital inflows can only cover a fraction of gross external financing requirements. Because there are no commitments of concessional financing or debt relief, for illustrative purposes, external financing gaps are identified and assumed to be financed on highly concessional terms. The scenario also assumes some real effective exchange rate appreciation reflecting positive productivity differentials with trading partners.

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[8] The DSA was undertaken by the Fund staff, in collaboration with the World Bank staff. It uses the standard Bank/Fund templates and sensitivity analyses for low-income countries (LICs) to assess Zimbabwe's debt dynamics.

[9] Staff estimates of the stock of external PPG medium- and long-term debt include (i) external PPG medium- and long-term debt based on original maturities; (ii) domestic foreign currency-denominated debt; (iii) arrears on intial principal and interest obligations; and (iv) estimated accrued penalties on arrears.

50. **Under the baseline scenario, PPG external debt indicators are projected to remain far in excess of the corresponding thresholds for a LIC that is a poor performer.**[10] These debt ratios begin at levels some three to five times above the thresholds. They would decline only gradually and approach the threshold levels after 20 to 30 years, assuming that external gaps are filled with concenssional financing (as mentioned above).

51. **The sensitivity analysis demonstrates an extreme fragility of Zimbabwe's external debt position.** Reflecting the poor past macroeconomic performance, under the historical scenario, all external debt indicators are projected to increase continuously over the long term.

Public debt sustainability

52. **The baseline scenario assumes improvements in fiscal policy implementation.** Budget revenue is projected to increase from 25 percent of GDP in 2009 to 30 percent in 2014 and beyond on account of better tax policy and administration. While the wage bill is assumed to be contained, large humanitarian and infrastructure rehabilitation needs would justify a significant increase in expenditure over the medium term. As a result, projected budget gross financing requirements are large over the medium term, but they are expected to decline over the longer term. For illustrative purposes, unidentified financing is reflected in budget financing gaps.

53. **The starting level of domestic government debt is moderate, and prudent domestic debt management policy is assumed going forward.** Local currency-denominated domestic debt was fully repaid in late January 2009. The RBZ's foreign currency-denominated domestic debt (which is added to total domestic public debt for the purpose of this DSA) is estimated at US$457 million (13 percent of GDP) at end-December 2008. The baseline scenario assumes modest government domestic borrowing in 2009 and over the medium term. As a result, domestic government debt-to-GDP ratios are projected to decline gradually over the medium and long term.

54. **However, under the baseline scenario, Zimbabwe's overall public debt**[11] **is unsustainable.** The initial external public debt overhang and large budget financing requirements would prevent public external debt ratios from reaching sustainable levels in the medium and even long term. Public debt is projected to gradually decline from a peak of over 200 percent of GDP in 2010 to about 150 percent a decade later. The PV of the public

[10] Zimbabwe is treated as a weak policy performer for the specific purposes of this LIC DSA. The policy-based thresholds for the present value (PV) of PPG external debt are therefore low: 200 percent of revenue; 100 percent of exports; and 30 percent of GDP.

[11] Including the domestic and external debt of both central government and parastatals.

debt-to-revenue ratio remains above 400 percent through 2018; and the debt service-to-revenue ratio only falls below 20 percent in 2014.

Table 7. External Debt Sustainability Framework, Baseline Scenario, 2006–29 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical	Standard	Projections						Average	Projections		Average
	2006	2007	2008	Average	Deviation	2009	2010	2011	2012	2013	2014	2009–14	2019	2029	2015–29
External debt (nominal) 1/	**120.5**	**139.6**	**174.8**			**207.0**	**211.0**	**210.2**	**210.0**	**207.1**	**202.3**		**177.1**	**93.0**	
Of which: public and publicly guaranteed (PPG)	117.9	136.6	166.0			197.0	199.9	197.9	196.5	192.3	188.2		165.4	84.6	
Change in external debt	11.9	19.1	35.1			32.3	4.0	-0.8	-0.2	-2.9	-4.9		-6.3	-8.1	
Identified net debt-creating flows	13.6	13.2	41.2			-1.1	-3.8	-7.7	-10.0	-9.0	-7.5		-8.1	-9.0	
Noninterest current account deficit	**4.6**	**2.3**	**23.7**	**6.4**	**8.1**	**14.2**	**25.5**	**22.2**	**18.6**	**16.3**	**14.9**		**9.7**	**4.2**	**7.8**
Deficit in balance of goods and services	13.2	15.5	20.0			35.6	35.5	27.8	20.4	16.0	14.7		9.9	4.4	
Exports	52.7	56.3	57.2			50.3	46.8	47.9	47.9	47.5	46.0		43.1	38.9	
Imports	65.9	71.8	77.2			85.9	82.4	75.7	68.3	63.5	60.7		52.9	43.3	
Net current transfers (negative = inflow)	-11.8	-12.4	-20.0	-8.1	5.6	-27.8	-15.8	-11.4	-7.6	-5.5	-5.5		-5.5	-4.5	-5.3
Of which: official	-1.5	-1.2	-2.3			-2.6	-2.9	-2.5	-2.2	-2.0	-2.0		-2.0	-2.0	
Other current account flows (negative = net inflow)	3.2	-0.8	23.8			6.4	5.8	5.8	5.8	5.8	5.7		5.3	4.3	
Net FDI (negative = inflow)	**-1.1**	**-1.9**	**-1.4**	**-0.9**	**0.9**	**-1.7**	**-2.2**	**-3.0**	**-3.3**	**-3.4**	**-3.6**		**-4.7**	**-6.0**	**-5.1**
Endogenous debt dynamics 2/	**10.1**	**12.8**	**18.9**			**-13.6**	**-27.1**	**-26.9**	**-25.3**	**-21.9**	**-18.8**		**-13.1**	**-7.2**	
Denominator: 1+g+r+gr	1.0	1.0	0.9			1.1	1.2	1.2	1.1	1.1	1.1		1.1	1.1	
Contribution from nominal interest rate	6.4	7.4	2.5			2.3	2.2	2.0	1.6	1.5	1.5		1.3	0.7	
Contribution from real GDP growth	7.1	8.7	22.0			-4.5	-10.7	-11.0	-11.0	-11.2	-11.2		-9.3	-5.1	
Contribution from price and exchange rate changes	-3.4	-3.3	-5.7			-11.3	-18.6	-17.9	-16.0	-12.2	-9.0		-5.1	-2.8	
Residual (3-4) 3/	**-1.7**	**5.9**	**-6.1**			**33.3**	**7.7**	**6.9**	**9.8**	**6.1**	**2.6**		**1.8**	**0.9**	
Of which; exceptional financing	-5.9	-1.4	-12.9			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	191.9			179.9	168.3	157.6	148.6	142.9	137.5		118.2	67.1	
In percent of exports	...	...	335.5			357.8	359.4	328.8	310.0	301.1	298.6		274.4	172.6	
PV of PPG external debt	**...**	**...**	**183.1**			**169.9**	**157.2**	**145.3**	**135.1**	**128.1**	**123.5**		**106.5**	**58.7**	
In percent of exports	**...**	**...**	**320.1**			**337.9**	**335.8**	**303.2**	**281.8**	**269.9**	**268.2**		**247.4**	**151.0**	
In percent of government revenues	**...**	**...**	**4,378.3**			**675.3**	**575.2**	**518.3**	**465.4**	**420.3**	**405.0**		**349.4**	**192.7**	
Debt service-to-exports ratio (in percent)	**24.5**	**26.4**	**22.2**			**18.5**	**25.9**	**13.2**	**10.6**	**7.3**	**5.3**		**4.1**	**5.6**	
PPG debt service-to-exports ratio (in percent)	**22.8**	**24.9**	**19.7**			**14.2**	**16.3**	**12.7**	**10.2**	**6.8**	**4.8**		**3.7**	**5.1**	
PPG debt service-to-revenue ratio (in percent)	**85.8**	**247.2**	**270.1**			**28.4**	**27.9**	**21.7**	**16.8**	**10.6**	**7.3**		**5.2**	**6.6**	
Total gross financing need (Millions of U.S. dollars)	760	727	1,406			1,382	1,942	1,832	1,863	1,903	2,005		2,031	2,127	
Noninterest current account deficit that stabilizes debt ratio	-7.3	-16.8	-11.4			-18.0	21.6	23.0	18.8	19.2	19.7		16.0	12.3	
Key macroeconomic assumptions															
Real GDP growth (in percent)	-6.3	-6.9	-14.1	-6.3	3.6	2.8	6.0	6.0	6.0	6.0	6.0	5.5	5.5	5.5	5.5
GDP deflator in US dollar terms (change in percent)	3.2	2.9	4.2	5.7	42.8	6.9	9.8	9.3	8.2	6.2	4.6	7.5	2.8	2.8	3.0
Effective interest rate (percent) 5/	5.7	5.9	1.6	4.6	1.2	1.4	1.2	1.1	0.9	0.8	0.8	1.0	0.8	0.8	0.8
Growth of exports of G&S (US dollar terms, in percent)	1.3	2.4	-9.1	-3.1	6.7	-3.3	8.4	18.6	14.7	11.4	7.5	9.6	7.5	7.0	7.4
Growth of imports of G&S (US dollar terms, in percent)	1.3	4.3	-3.8	-3.3	10.5	22.4	11.7	6.6	3.5	4.5	6.0	9.1	6.0	7.0	6.2
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	61.0	61.0	61.0	61.0	61.0	61.0	61.0	61.0	61.0	61.0
Government revenues (excluding grants, in percent of GDP)	14.0	5.7	4.2			25.2	27.3	28.0	29.0	30.5	30.5		30.5	30.5	30.5
Aid flows (in Millions of US dollars) 7/	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Of which: Grants	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Of which: Concessional loans	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			20.0	18.6	15.5	12.4	10.0	8.4		4.1	0.2	3.0
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			61.0	61.0	61.0	61.0	61.0	61.0		61.0	61.0	61.0
Memorandum items:															
Nominal GDP (Millions of US dollars)	3,711	3,553	3,180			3,498	4,073	4,719	5,412	6,090	6,750		10,360	23,424	
Nominal dollar GDP growth	-3.3	-4.3	-10.5			10.0	16.4	15.9	14.7	12.5	10.8	13.4	8.5	8.5	8.6
PV of PPG external debt (in Millions of US dollars)			5,823			5,943	6,403	6,857	7,311	7,804	8,334		11,037	13,759	
(PVt-PVt-1)/GDPt-1 (in percent)						3.8	13.2	11.1	9.6	9.1	8.7	9.3	5.5	0.3	3.7

Source: Staff simulations.

1/ Includes both public and private sector medium- and long-term external debt; and foreign currency-denominated domestic public debt as from 2009.

2/ Derived as [r - g - ρ(1+g)]/(1+g+r+gr) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value. Foreign currency-denominated domestic public debt is not included in PV calculations.

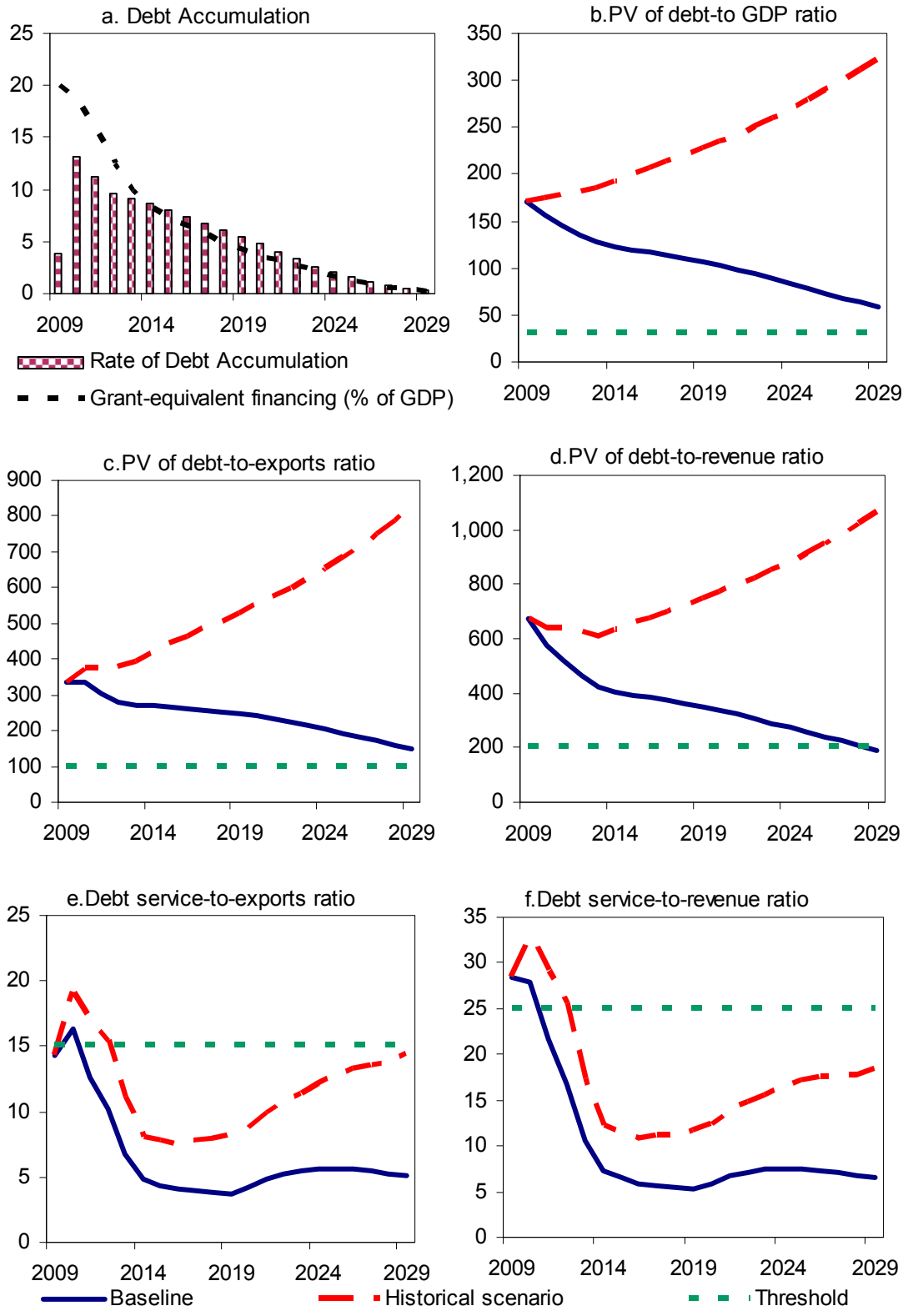
5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

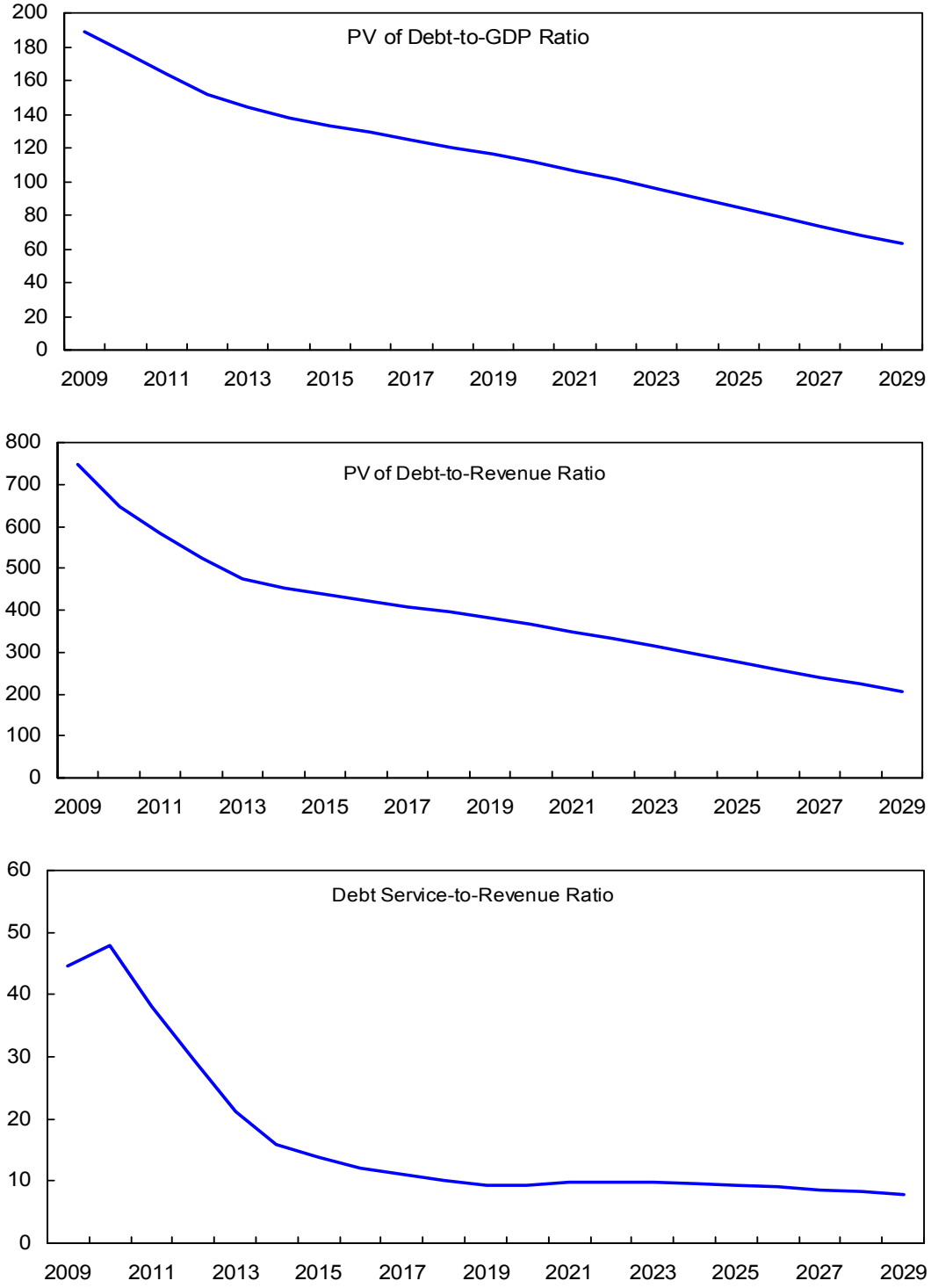
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Figure 2. Zimbabwe: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2009–29



Source: IMF staff projections and simulations.

Figure 3. Zimbabwe: Indicators of Public Debt, Baseline Scenario, 2009–29 1/



Source: IMF staff projections and simulations.

1/ Estimates of the stock of public debt in 2009 include central government, RBZ, parastatal and publicly-guaranteed liabilities. Forecasts of revenue and expenditure are for central government only.